

ABCA ACTIVITY DOCUMENT

Cryptography – Laboratory Activity

Course Name	Cryptography
Course Code	CSL0509
Semester	Cryptography (CSL0509) – B.Tech AIML V Semester
Faculty Name	Mr. Hirendra Singh Sengar
Academic Year	2026–27

1. Description of Activity

This activity requires students to perform hands-on laboratory experiments in the domain of Cryptography and Network Security using industry-standard tools such as CrypTool, OpenSSL, Visual Studio Code, Wireshark, Kali Linux, and Cisco Packet Tracer. Students are expected to solve the assigned practical problem statements, document their working procedure, and deploy the complete lab record on their personal GitHub repository.

2. Objectives

1. To apply classical and modern cryptographic algorithms for data encryption and decryption.
2. To analyze the strength, vulnerabilities, and performance of various cryptographic techniques.
3. To gain hands-on exposure to real-world security tools used in industry and research.
4. To develop the practice of version-controlled documentation using GitHub.

3. Bloom's Taxonomy Levels Covered

Level	Description
BL3	Apply
BL4	Analyze

4. Instructions to Students

1. Solve the practical problem statements listed below using the specified tools.
2. For each experiment, upload the following to your GitHub repository: working procedure of the experiment, screenshot(s) of the tool setup/environment, algorithm (if applicable), and screenshot(s) of the output/result.
3. Maintain a separate, clearly labeled folder for each experiment (e.g., Exp1_CaesarCipher, Exp2_FrequencyAnalysis, etc.).
4. Share the GitHub repository link as instructed by the faculty on or before the due date.
5. Plagiarism or copied repositories will be evaluated as per institute norms.

5. Marks Distribution

Component	Marks
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Lab Experiments (GitHub Deployment – Problem Statements)	40
Total	40

6. List of Practical Problem Statements

S.No.	Practical Problem Statement	Tool Required	BL3–BL4
1	Implement Caesar, Vigenère, and Rail Fence ciphers to encrypt and decrypt plaintext, and compare their working and effectiveness. The given plain text is “HELLO WORLD”.	CrypTool	BL4 (Analyze)
2	Perform frequency analysis on classical cipher texts and analyze their vulnerability to cryptanalysis attacks using CrypTool.	CrypTool	BL4 (Analyze)
3	Encrypt sample data using DES and AES algorithms, and compare their security features and performance.	CrypTool	BL4 (Analyze)
4	Generate RSA public and private key pairs and perform secure file encryption and decryption using OpenSSL.	OpenSSL	BL3 (Apply)
5	Generate SHA-256 hash values, create digital signatures, and verify their authenticity using OpenSSL.	OpenSSL	BL3 (Apply)
6	Create a self-signed X.509 digital certificate and analyze its attributes, validity, and security features.	OpenSSL	BL4 (Analyze)
7	Implement RSA and Diffie–Hellman algorithms to demonstrate secure key exchange and public-key cryptography using VS Code.	Visual Studio Code (VS Code)	BL3 (Apply)
8	Develop an application to secure text messages using AES encryption and SHA-256 hashing using VS Code.	Visual Studio Code (VS Code)	BL3 (Apply)
9	Implement password hashing with and without salting, and evaluate resistance against common password attacks using VS Code.	Visual Studio Code (VS Code)	BL4 (Analyze)
10	Capture HTTPS network traffic using Wireshark and analyze the TLS handshake, digital certificates, and cipher suites.	Wireshark	BL4 (Analyze)
11	Capture and analyze encrypted VPN traffic using Wireshark, and compare it with unencrypted network communication.	Wireshark	BL4 (Analyze)
12	Analyze captured network packets to identify secure and insecure communication protocols and their security implications.	Wireshark	BL4 (Analyze)
13	Perform password auditing using John the Ripper and evaluate password strength against dictionary and brute-force attacks.	Kali Linux	BL4 (Analyze)
14	Generate SSH/RSA key pairs and configure passwordless authentication between Linux systems.	Kali Linux	BL3 (Apply)
15	Configure a secure VPN using OpenVPN and verify encrypted communication between connected systems.	Kali Linux	BL3 (Apply)
16	Design and implement a secure wireless network using WPA2 encryption (AES) in Cisco Packet Tracer, and verify secure communication.	Cisco Packet Tracer	BL3 (Apply)

7. Submission Checklist

- ☐ GitHub repository created and shared
- ☐ All 16 experiments documented in separate folders

- ☐ Working procedures included for each
- ☐ Screenshots of tool setup included
- ☐ Screenshots of outputs included
- ☐ Algorithms included wherever applicable
- ☐ Repository link submitted before deadline